

Class: 6

Time: 90 Minutes

Maximum Marks: 100

General Instructions:

1. This paper is divided into four sections: A, B, C, and D.
 2. **Section A** consists of 25 Multiple Choice Questions (MCQs). Each question carries 2 marks. ($25 \times 2 = 50$ marks)
 3. **Section B** consists of 10 True/False and Fill in the Blanks questions. Each question carries 1 mark. ($10 \times 1 = 10$ marks)
 4. **Section C** consists of 5 Matching Type questions. Each question carries 2 marks. ($5 \times 2 = 10$ marks)
 5. **Section D** consists of 10 Short Answer and Case-Based questions. Each question carries 3 marks. ($10 \times 3 = 30$ marks)
 6. All questions are compulsory. Write neatly and number your answers correctly.
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Section A: Multiple Choice Questions ($25 \times 2 = 50$ Marks)

1. A potted plant is placed inside a cardboard box with a small hole on one side. After two weeks, the plant's stem is observed bending towards the hole. This experiment demonstrates:

- a) Photosynthesis
- b) Phototropism
- c) Geotropism
- d) Hydrotropism

2. Which of the following materials is a conductor of electricity but NOT a conductor of heat?

- a) Copper
- b) Iron
- c) Aluminum
- d) All conductors of electricity are also conductors of heat

3. A student dissolves sugar in water. Which of the following statements is INCORRECT?

- a) Sugar is the solute and water is the solvent
- b) The sugar particles spread evenly throughout the water
- c) The sugar can be separated by filtration
- d) The mixture is called a solution

4. The process of converting animal waste and dead leaves into manure by the action of microorganisms is called:

- a) Fermentation
- b) Pasteurization

- c) Composting
- d) Distillation

5. A marble is rolled on four different surfaces: polished marble floor, sand, wet grass, and a carpet. On which surface will the marble travel the FURTHEST before stopping?

- a) Sand
- b) Carpet
- c) Wet grass
- d) Polished marble floor

6. Which of the following statements about the Moon is TRUE?

- a) The Moon produces its own light
- b) The Moon's gravity is stronger than Earth's gravity
- c) The Moon rotates on its axis and revolves around the Earth in the same time
- d) The Moon has an atmosphere like Earth

7. A paper clip is attracted to a magnet. If the paper clip is heated strongly and then brought near the magnet again, what will most likely happen?

- a) It will be attracted more strongly
- b) It will be attracted less strongly or not at all
- c) It will be repelled by the magnet
- d) It will become a permanent magnet

8. During digestion, proteins are broken down into:

- a) Fatty acids
- b) Glucose
- c) Amino acids
- d) Starch

9. Which of the following is a periodic change?

- a) Rusting of iron
- b) Burning of paper
- c) Change of seasons
- d) Melting of ice cream

10. A student wants to measure the volume of an irregularly shaped stone. Which apparatus would be MOST suitable?

- a) Measuring cylinder and water
- b) Ruler and scale
- c) Weighing balance
- d) Thermometer

11. Which vitamin is produced in our body when our skin is exposed to sunlight?

- a) Vitamin A
- b) Vitamin B complex
- c) Vitamin C
- d) Vitamin D

12. The stem of a cactus plant becomes thick and fleshy to:

- a) Attract insects for pollination
- b) Store water for dry periods
- c) Protect the plant from animals
- d) Absorb more sunlight

13. A shadow is shortest when the sun is:

- a) Rising in the east
- b) Setting in the west
- c) Directly overhead at noon
- d) Just below the horizon

14. Which of the following is a reversible change?

- a) Baking a cake
- b) Ripening of fruits
- c) Melting of wax
- d) Curdling of milk

15. The unit used to measure the loudness of sound is:

- a) Hertz (Hz)
- b) Decibel (dB)
- c) Newton (N)
- d) Joule (J)

16. A boat floats on water because:

- a) The boat is lighter than water
- b) The boat is made of wood
- c) The water pushes the boat upward with a force greater than the boat's weight
- d) The boat's shape allows it to displace water whose weight is equal to the boat's weight

17. Which of the following is NOT a characteristic of a desert animal like a camel?

- a) Long eyelashes to protect from sand
- b) Ability to store fat in humps
- c) Thick fur to keep warm in cold weather
- d) Wide feet to walk on sand easily

18. The process by which plants lose excess water through their leaves is called:

- a) Transpiration
- b) Respiration
- c) Photosynthesis
- d) Condensation

19. A bulb does NOT glow in a circuit. Which of the following could be the reason?

- a) The bulb is fused
- b) The wires are not connected properly
- c) The cells are exhausted
- d) All of the above

20. Which of these is a translucent material?

- a) Clear glass
- b) Wooden board
- c) Frosted glass
- d) Steel plate

21. The joint between the bones of the skull is an example of a:

- a) Hinge joint
- b) Fixed joint
- c) Pivot joint
- d) Ball and socket joint

22. Water in a lake evaporates on a hot day. The water vapor rises and forms clouds. This is an example of:

- a) The water cycle
- b) A chemical change
- c) A reversible chemical reaction
- d) Pollution

23. Which organ in the human body is responsible for filtering waste from the blood?

- a) Heart
- b) Lungs
- c) Kidney
- d) Stomach

24. A student classified materials as magnetic and non-magnetic. Which of the following pairs is classified INCORRECTLY?

- a) Iron nail – Magnetic
- b) Copper wire – Non-magnetic
- c) Stainless steel spoon – Magnetic
- d) Plastic ruler – Non-magnetic

25. Which statement best explains why we see lightning before we hear thunder?

- a) Lightning occurs before thunder
- b) Light travels faster than sound
- c) Sound travels faster than light
- d) Lightning is closer to us than thunder

Section B: True/False and Fill in the Blanks (10 × 1 = 10 Marks)

A. True or False (Write 'True' or 'False')

26. All insects have six legs and three body parts.

27. The Earth's rotation on its axis causes the change in seasons.

28. A mixture of sand and water can be separated by filtration.
29. The heart is made up of cardiac muscles that work involuntarily.
30. Non-renewable sources of energy like coal and petroleum can be replenished quickly.

B. Fill in the Blanks (Write the complete sentence with the correct word)

31. The process of a solid changing directly into a gas without becoming a liquid is called _____.
32. The smallest unit of living things is the _____.
33. A _____ is a device used to break or complete an electric circuit.
34. The force that pulls objects towards the center of the Earth is called _____.
35. The green pigment in plants that helps in photosynthesis is called _____.
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Section C: Matching Type Questions ($5 \times 2 = 10$ Marks)

Match Column A with Column B. Write the correct letter next to the number.

Column A

36. Vitamin C
37. Vitamin A
38. Iodine deficiency
39. Iron deficiency
40. Vitamin D deficiency

Column B

- a. Goitre
- b. Rickets
- c. Night blindness
- d. Scurvy
- e. Anemia

Answers format: 36-d, 37-c, etc.

Section D: Short Answer and Case-Based Questions ($10 \times 3 = 30$ Marks)

41. The Mystery of the Disappearing Water (Case-Based)

Rahul poured 100 ml of water into a shallow dish and 100 ml of water into a narrow, deep bottle. He kept both containers on the same windowsill. After three days, he observed that more water had evaporated from the dish than from the bottle.

- a) What is evaporation? (1 mark)
- b) Why did more water evaporate from the shallow dish? (1 mark)
- c) Name one other factor that could increase the rate of evaporation. (1 mark)

42. The Curious Case of the Sinking Orange (Application-Based)

A whole orange with its peel sinks when placed in water. However, when the peel is removed, the peeled orange sinks even faster.

- a) Why does an orange float better with its peel on? (1 mark)
- b) What does this tell you about the density of the peel compared to water? (1 mark)
- c) Name one other fruit or object that behaves similarly. (1 mark)

43. The Bouncing Ball Experiment (Experimental Design)

A student drops a rubber ball from a height of 100 cm and measures how high it bounces. She repeats the experiment with the same ball from different heights.

- a) Identify the independent variable in this experiment. (1 mark)
- b) Identify the dependent variable. (1 mark)
- c) What hypothesis could the student be testing? (1 mark)

44. The Wonders of the Human Body (Conceptual Understanding)

- a) Why do our muscles work in pairs? Explain with an example. (2 marks)
- b) What is the function of the rib cage? (1 mark)

45. The Disappearing Sugar (Chemical vs. Physical Change)

Ravi dissolved sugar in a glass of water and noticed the sugar disappeared. His friend said it was a chemical change.

- a) Is dissolving sugar a physical or chemical change? (1 mark)
- b) Give one reason to support your answer. (1 mark)
- c) How can Ravi get the sugar back from the solution? (1 mark)

46. The Magnetic Mystery (Logical Reasoning)

A student has three metal rods labeled A, B, and C. Rod A attracts both Rod B and Rod C. Rod B and Rod C repel each other.

- a) Which rod is definitely a magnet? (1 mark)
- b) What can you conclude about Rod B and Rod C? (1 mark)
- c) If Rod A is brought near a compass, what would you observe? (1 mark)

47. The Secret of the Shadows (Light and Shadows)

A student observes her shadow in the morning and again at noon. She notices her shadow is very long in the morning but very short at noon.

- a) Why is the shadow longer in the morning? (1 mark)
- b) What will happen to the shadow in the evening? (1 mark)
- c) Name one object that does not form a shadow. (1 mark)

48. The World of Fibres (Material Science)

Cotton and polyester are two fabrics used to make clothes.

- a) Which of these is a natural fibre and which is synthetic? (1 mark)
- b) Why is cotton preferred for making summer clothes? (1 mark)
- c) Why might a raincoat be made of polyester or nylon instead of cotton? (1 mark)

49. The Water Cycle in Action (Environmental Science)

- a) What is condensation in the water cycle? (1 mark)
- b) How are clouds formed? (1 mark)
- c) Name two forms of precipitation. (1 mark)

50. The Plant's Amazing Journey (Life Processes)

A seed is planted in soil and kept in a dark room. After a few days, a small shoot appears, but the leaves are yellow.

- a) Why did the leaves turn yellow instead of green? (1 mark)
- b) What happens to the plant if it is moved to sunlight? (1 mark)
- c) What is the process by which plants make their own food? (1 mark)